

$\phi: \Sigma$ (spacetime) $\longrightarrow M$ (target)

(i) $\Sigma \cong \mathbb{M}^n$ field theory proper / dynamic

(ii) $\Sigma \cong \mathbb{R}^n$: static equilibrium situations

$$S(\phi) = \int_{\Sigma} |\mathrm{d}\phi|^2 \mathrm{d}v$$

$$\bullet \delta S(\phi_{\text{cl}}) = 0$$

$$\hookrightarrow \text{Feynmann: } \int_{\text{Map}(\Sigma, M)} F(\phi) e^{-\frac{i}{\hbar} S(\phi)} \mathcal{D}\phi$$

$F(\phi)$: "insertion" onto the spacetime

$\mathcal{D}\phi \equiv \prod_{x \in \Sigma} \mathrm{d}\mu(\phi(x))$ (Measure on M).

$\bullet$ Each ϕ contributes with a probability

$$p(\phi) \approx e^{-\frac{i}{\hbar} S(\phi)}$$

* For $\mathbb{M}^n$ these are not probabilities since $S(\phi) \in \mathbb{C}$.

Eulerian free field Gaussian:

$\text{Map}(\Sigma, M)$: (i) vector space inherited from M .

(ii) Union of affine spaces $\Rightarrow S \propto \phi^2$

* Let (Σ, χ) be a Riemannian (d+1)-dim. oriented

compact mfd with $M = \mathbb{R}$
 free field theory as an L^2 inner product on Σ
 $dV \equiv$ Riemannian volume integral
 $\frac{\beta}{2u} G \equiv (-\Delta + m^2)^{-1}$ (propagator)

$$\begin{aligned} S &= \int dV (\partial_\mu \phi \partial^\mu \phi + m^2 \phi^2) \\ &= \frac{\beta}{2u} \int dV \phi (-\Delta + m^2) \phi \\ &= \frac{1}{2} \int dV \phi (G^{-1} \phi) \\ &\equiv \frac{1}{2} (\phi, G^{-1} \phi)_{L^2} \end{aligned}$$

Partition f^u :

$$Z[\phi] = \int_{\text{Map}(\Sigma, \mathbb{R})} e^{-S(\phi)} D\phi = \int_{\text{Map}(\Sigma, \mathbb{R})} e^{-\frac{1}{2}(\phi, G^{-1} \phi)} D\phi$$

$\text{Map}(\Sigma, \mathbb{R})$: Hilbert space with L^2 inner product
 $(\cdot, \cdot) : \mathcal{H} \times \mathcal{H} \rightarrow \mathbb{R}$

For a finite dimensional space $\text{Map}(\Sigma, \mathbb{R})$

$$\int \prod_i \frac{d\phi_i}{2\pi} e^{-\frac{1}{2}(\phi, G^{-1} \phi)}$$

Expand ϕ in $\text{Map}(\Sigma, \mathbb{R})$ in the basis of orthonormal maps associated to G , i.e. over its spectrum
 $\phi = \lambda_i \phi_i$

$$\begin{aligned} \Rightarrow \int \prod_i \frac{d\phi_i}{2\pi} e^{-\frac{1}{2}(\lambda_i \phi_i, G^{-1} \lambda_j \phi_j)} &= \int \prod_i \frac{d\phi_i}{2\pi} e^{-\frac{1}{2}(\phi, G^{-1} \phi)} \\ &= \int \prod_i \frac{d\phi_i}{2\pi} e^{-\frac{1}{2} \lambda_i (\phi_i, \phi_i)} = \int \prod_i \frac{d\phi_i}{2\pi} e^{-\frac{1}{2} \lambda_i \int \phi_i^2 dV} \end{aligned}$$

$$= \int \prod_i \frac{d\phi_i}{2\pi} (1) = \frac{1}{2\pi} \sqrt{\frac{\pi}{\lambda_i}}$$

$$\int d\bar{V} e^{-\bar{V}^T A \bar{V}}$$

diagonalize

$$V = U \mathcal{U}$$

$$V^T = \mathcal{U}^T U^T$$

$\Rightarrow$ then we have $(\phi, G^{-1} \phi)$

$$\phi = A\psi$$

$$(A\psi, G^{-1}A\psi) = (\psi, A^T G^{-1} A \psi)$$

$$\text{s.t. } A^T G^{-1} A \psi = \lambda \psi$$

$$\begin{aligned} \bullet d\phi_i &= d(A\psi) = \\ &= d(A^T_\mu \psi_\nu)_i = (A^T_\mu d\psi_\nu)_i \\ \Rightarrow A^T_\mu (d\psi_i)_\nu &\sim \dots \sim \lambda (d\psi)_i \\ &= \det(A) d\psi_i \dots, \text{ hence } \det(A) = 1 \end{aligned}$$

so that

$$\begin{aligned} \pi \int \frac{d\psi_k}{\sqrt{2u}} e^{-\frac{1}{2}\lambda(\psi, \psi)} &= \pi \int \frac{d\psi_k}{\sqrt{2u}} e^{-\frac{1}{2}\psi_k^2} \\ &= \frac{\pi}{\sqrt{2u}} \sqrt{\frac{2\pi}{\lambda_k}} = \sqrt{\frac{\pi}{\lambda_k}} = \sqrt{\det(G^{-1})^{-1}} \end{aligned}$$

$$\det(G^{-1}) = \det(G)^{-1}$$

$$\therefore \boxed{Z = (\det(G))^{1/2}}$$

$$\ast \text{ } (\mathbb{R}/2\pi)G = (-\Delta + m^2)^{-2} \lambda_n, n=1, 2, \dots$$

behaves as $\mathcal{O}(n^{-2/(d+1)})$

$\bullet G$ is a positive operator, hence $\lambda_n, n \geq 0$

$\ast$ For $d=4$: $\mathcal{O}(n^{-2/5})$, use zeta-fⁿ regularization

$$\boxed{\det(G) \equiv \exp(-\zeta'_G(0))}$$

$$\zeta_G(s) \equiv \sum_n \lambda_n^{-s}, \text{ Re}(s) < -\frac{d+1}{2}$$

$$\rightarrow \zeta'_G(s) = \sum_n \frac{d}{ds} \lambda_n^{-s} \ln(\lambda_n) = \sum_n -\ln(\lambda_n) \lambda_n^{-s}$$

$$\zeta'_G(0) = \sum_n \ln(\lambda_n) = -\sum_n \ln(\lambda_n)$$

$\det(G)$

$$\begin{aligned} \ast \text{ Feynman-Kac formula: } \phi &\equiv \sqrt{\frac{u}{8m}} (a + a^\dagger) \\ \int_{\Gamma_u[0,1]} \phi(x_1) \dots \phi(x_n) d\mu_G(\phi) &= \text{Tr}(e^{-x_1 H} \phi e^{(x_1 - x_2) H} \phi \dots \phi e^{(x_{n-1} - 1) H}) \end{aligned}$$

$$\bullet \text{ Ground state } |\Omega\rangle \propto e^{-\frac{8m}{4u}\phi^2}$$

$$\text{Prob 1: } \int_{\Gamma_u[0,1]} \phi(x_1) \dots \phi(x_n) d\mu_G(\phi) = \dots$$

$$Z_{pt} \int_{C_{pt}[0,1]} \phi(x_1) \phi(x_2) d\mu_G(\phi) = \frac{\text{Tr}(e^{-x_1 H} \phi e^{(x_2-1)H})}{\text{Tr}(e^{-LH})}$$

Diffeomorphism covariance: (Σ, γ)

$$Z_\gamma = Z_{D^*\gamma}$$

$$\Rightarrow \langle \phi_{\ell_1}(D(x_1)) \dots \phi_{\ell_n}(D(x_n)) \rangle_\gamma = \langle \phi_{\ell_1}(x_1) \dots \phi_{\ell_n}(x_n) \rangle_{D^*\gamma}$$

$$\begin{array}{ccc} T_{(0,2)}^* \Sigma & \xleftarrow{D^*\gamma} & T_{(0,2)}^* \Sigma \\ \downarrow \pi & & \downarrow \pi \\ \Sigma & \xrightarrow{D} & \Sigma \end{array}$$

$\langle \phi_{\ell_1}(D(x_1)) \dots \phi_{\ell_n}(D(x_n)) \rangle_\gamma$: uses (Σ, γ)
 and so looking at $\langle \phi_{\ell_1}(x_1) \dots \phi_{\ell_n}(x_n) \rangle$ on Σ
 we have $\pi \circ D = D^*\gamma \circ \pi^{-1}$

(ii) local scale covariance: (Weyl)

$$Z_{g\gamma} = \exp \left[\frac{c}{96\pi} \left(\|d\sigma\|_{L^2}^2 + 4 \int_\Sigma \sigma r dv \right) \right] Z_\gamma$$

$$\|d\sigma\|_{L^2}^2 = \int_\Sigma (\partial^\nu \sigma) \partial_\nu \sigma \gamma_{\mu\nu} dv$$